SHAHZAD ALI

Data Science · Graph Machine Learning · Brain Connectomics · Multimodal Neuroimaging · Explainable AI

PhD candidate in Data Science and Computation (thesis submitted Oct 2026; defence expected Mar–Apr 2027)
Alma Mater Studiorum – Università di Bologna, Italy · LISCOMP Lab, IRCCS San Martino, Genoa, Italy
Genoa, Italy
shahzadali039@gmail.com · +39 348 265 7723
Portfolio · GitHub · LinkedIn · ORCID · Google Scholar · ResearchGate · Scopus

Research Profile

Machine learning researcher developing graph representation learning and multimodal learning methods for neuroimaging, with a focus on Alzheimer's disease diagnosis, staging and progression modelling. My work covers individualised and multiplex brain connectomes, hierarchical graph neural networks (GNNs), learning from incomplete imaging and fluid-biomarker profiles, uncertainty and calibration, and explainability tested against autopsy-confirmed neuropathology, with leakage-safe evaluation and reproducible pipelines. Doctoral research at the LISCOMP Laboratory (MIDA group, University of Genoa × IRCCS San Martino), with a research visit to the LILI Lab, University of Sussex.

Education

11/2022 – present **PhD in Data Science and Computation**, Alma Mater Studiorum – Università di Bologna, Italy. Thesis submitted 10/2026; defence expected 03–04/2027.
Graph Neural Networks and Multimodal Learning for Alzheimer's Disease: From Prediction to Pathological Attribution.
Funded by Next Generation EU – NRRP (D.M. 117/2023).

07/2025 – 01/2026 **Visiting Researcher (PhD mobility)**, LILI Lab, Sussex AI Centre, University of Sussex, UK.

01/2016 – 07/2018 **MSc in Computer Science** (CGPA 3.75/4.00), University of Engineering and Technology (UET), Lahore, Pakistan. Thesis on neural-network rainfall–runoff forecasting; PEEF talent scholarship.

08/2011 – 06/2015 **BSc in Telecommunication and Networking** (CGPA 3.46/4.00), COMSATS University Islamabad, Pakistan. ICT R&D merit scholarship.

Research Experience

09/2023 – present **PhD Researcher**, LISCOMP Lab, IRCCS Ospedale Policlinico San Martino, Genoa, Italy

- Built pipelines from raw structural MRI, diffusion MRI and PET to individualised structural connectomes and regional features (FreeSurfer, MRtrix3, Clinica) on ADNI and NACC.
- Developed and benchmarked ensemble, GNN and uncertainty-aware multimodal fusion models for AD staging and progression (sMCI vs. pMCI), with nested cross-validation, held-out testing, ablation and leakage-safe normalisation.
- Software, formal analysis and interpretation for collaborative clinical studies on tau staging, neurotransmitter and vascular correlates of AD, and cerebral amyloid angiopathy.

07/2025 – 01/2026 **Visiting Researcher**, LILI Lab, University of Sussex, Brighton, UK

- Designed and evaluated Graph-of-Graphs, a hierarchical GNN that fuses modality-specific subject-level brain graphs and connects subjects in a learned population graph, for AD staging and progression (first-author manuscript under review).

Selected Contributions

1. **Learning from incomplete multimodal data.** Uncertainty-aware fusion predicting from any available subset of six imaging and biofluid sources without imputation; on 2,359 ADNI participants, the durable gain from additional evidence was calibration rather than accuracy (MLCN @ MICCAI 2026).

2. **Hierarchical graph learning.** Graph-of-Graphs couples subject-level brain graphs to a population graph and substantially improved staging (under review).
3. **Connectomics and systematic evaluation.** On ADNI (394 subjects), morphometric plus microstructural features gave 0.898 balanced accuracy for CN vs. AD, while connectivity metrics added value only for some tasks (*Frontiers in Neuroimaging*, 2026).
4. **Neuropathological validity.** In 489 autopsy-confirmed NACC participants, grey-matter loss tracked Alzheimer pathology in 32 temporal and limbic measures, with no independent Lewy body association (under review).

Publications

12 peer-reviewed journal articles, 7 peer-reviewed conference papers and 3 first-author manuscripts under review; 310 citations, h-index 6 (Google Scholar, 25 October 2026). Full list: portfolio.

First-author research and reviews

- **Ali S**, Kreshpa W, Rosso N, Piana M, Roccatagliata L, Cirone A, Lorenzini L, Campi C, Pardini M, Garbarino S (2026). A systematic study on the integration of MRI connectivity metrics for Alzheimer's diagnosis, staging, and cognitive decline prediction. *Frontiers in Neuroimaging*, 5, 1746464. doi:10.3389/fnimg.2026.1746464
- **Ali S**, Garbarino S (2026). From Blood to Brain: Uncertainty-aware adaptive fusion for Alzheimer's staging and progression under incomplete multimodal profiles. MLCN 2026 @ MICCAI 2026, Strasbourg. Accepted, Springer LNCS.
- **Ali S**, Piana M, Pardini M, Garbarino S (2025). Graph neural networks in Alzheimer's disease diagnosis: a review of unimodal and multimodal advances. *Frontiers in Neuroscience*, 19, 1623141. doi:10.3389/fnins.2025.1623141

First-author manuscripts under review

- **Ali S**, Wijeratne P, Simpson I, Pardini M, Garbarino S. Graph-of-Graphs: Hierarchical multimodal graph learning for Alzheimer's disease staging and progression modeling. *Brain Informatics*.
- **Ali S**, Lorenzini L, Pardini M, Garbarino S. Explainability in graph neural networks for Alzheimer's disease neuroimaging: a scoping review of methods, biomarkers, and clinical reliability. *Artificial Intelligence Review*.
- **Ali S**, et al. Linking regional brain atrophy to its underlying pathology across the Alzheimer–Lewy spectrum. *Neurology*.

Collaborative research in Alzheimer's disease

- Pulze M, Garbarino S, Lorenzini L, Cirone A, **Ali S**, et al. (2026). Structural MRI signature predicts tau staging in Alzheimer's disease. *Alzheimer's & Dementia: DADM*, 18, e70451. doi:10.1002/dad2.70451
- Argenti L, Massa F, Losa M, . . . , **Ali S**, . . . , Pardini M (2026). Neurotransmitter landscape and neurodegeneration patterns in Alzheimer's disease. *Neurobiology of Aging*, 162, 49–56. doi:10.1016/j.neurobiolaging.2026.02.004
- Argenti L, Cirone A, Massa F, . . . , **Ali S**, . . . , Pardini M (2026). Exploring the spatial covariance of cerebral vascular density and amyloid burden in Alzheimer's disease. *Alzheimer's & Dementia: DADM*, 18(2), e70390. doi:10.1002/dad2.70390
- Losa M, Cotta Ramusino M, Cama I, . . . , **Ali S**, . . . , Pardini M (2025). Cerebrospinal fluid biomarkers profiling in cerebral amyloid angiopathy and relationship with disease phenotypes. *Journal of the American Heart Association*, 14(20), e044784. doi:10.1161/JAHA.125.044784

Machine learning and statistics in other domains

- Kiran I*, **Ali S***, et al. (2025). An AI-enabled framework for transparency and interpretability in cardiovascular disease risk prediction. *Computers, Materials & Continua*, 82(3). doi:10.32604/cmc.2025.058724 (*equal first authors)

- Shahbaz M, **Ali S**, et al. (2019). Classification of Alzheimer's disease using machine learning techniques. DATA 2019, SciTePress. doi:10.5220/0007949902960303
- **Ali S**, Shahbaz M (2020). Streamflow forecasting by modeling the rainfall–streamflow relationship using artificial neural networks. *Modeling Earth Systems and Environment*, 6(3). doi:10.1007/s40808-020-00780-3
- Further work: wind power forecasting (*PLOS ONE*, 2026); statistical analysis in three clinical dental studies (*Scientifica*, 2026; *Australian Endodontic Journal*, 2025; *J. Contemporary Clinical Practice*, 2026); five applied ML conference papers (three in Springer LNNS 2026; IEEE ICOSST 2019; YUENG 2019).

Technical Expertise

- **Graph and deep learning:** PyTorch, PyTorch Geometric; GCN, GAT, GraphSAGE, GIN; hierarchical and population-graph models; gated multimodal fusion.
- **Neuroimaging and connectomics:** FreeSurfer, MRtrix3, FSL, SPM, Clinica, Nilearn; individualised and multiplex connectome construction.
- **Evaluation, uncertainty and explainability:** nested and leakage-safe cross-validation, ablation, calibration, permutation testing, Optuna; SHAP, LIME, Grad-CAM, ElasticNet attribution.
- **Computing:** Python, scikit-learn, R, Git, Linux, LaTeX; documented pipelines on GitHub.

Teaching and Supervision

03/2019 – 10/2022	**Lecturer in Computer Science / IT**, University of Education, Lahore, Pakistan. Independently taught undergraduate Artificial Intelligence, Data Mining and Introduction to Machine Learning; supervised final-year projects.
10/2016 – 05/2018	**Graduate Research Assistant**, UET Lahore. Designed and taught the Artificial Intelligence laboratory; supervised class and final-year projects.

Presentations, Awards and Service

- **Presentations:** paper, MLCN @ MICCAI 2026 (Strasbourg); posters, OHBM 2026 (Bordeaux), LoG Italian Meetup 2026 (Pisa), RIN IRCCS Network 2024; invited speaker, National Centre for Physics, Islamabad (2019).
- **Funding and awards:** fully funded PhD scholarship, Next Generation EU – NRRP (2022); PEEF talent scholarship (2016); ICT R&D project grant and merit scholarship (2011–2015).
- **Training:** EEML Summer School (2023); Geilo Winter School on Graphs and Applications (2024); IEEE ComSoc eHealth PhD School (2023).
- **Peer review:** 12 reviews for *PLOS ONE*, *IEEE Access*, *Engineering Applications of Artificial Intelligence* and *Computers, Materials & Continua* (2018–2026).
- **Languages:** Urdu (native); English (fluent); Italian (basic).

References

Available upon request.